



ACROPOLIS[®]
Enlightening Wisdom



CSI AITR
Student Chapter

अनन्तः

24 Hours National Level Hackathon

ना ही जिसका कोड आदि और ना ही कोई अंत ।

PROBLEM STATEMENTS
2026

ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

1. Multimodal Crisis Intelligence Hub

Problem: Disaster data exists in scattered silos satellite feeds, social posts, sensors delaying critical rescue responses.

Description: Combines all disaster data into one real-time AI platform. Predicts crisis spread and instantly recommends actions for faster emergency decisions.

2. Predictive Healthcare & Agricultural Nexus

Problem: Crop failures and disease outbreaks hit without warning; farming and health data never connect until it's too late.

Description: Links weather, soil, crop, and health data in one system. Spots food and health risks early so communities can act before crises escalate.

3. Autonomous Construction Safety Ecosystem

Problem: Construction sites kill thousands yearly safety systems only react after accidents happen.

Description: Uses cameras, wearables, and machine sensors to predict dangers. Auto-triggers alerts or stops work before accidents occur.

4. Federated Learning for Environmental & Health Intelligence

Problem: Hospitals and farms hold valuable data but privacy rules prevent sharing, trapping insights in silos.

Description: Lets organizations train AI together without sharing raw data. Unlocks collective intelligence while keeping sensitive information private.

WEB DEVELOPMENT

1. AI Code Review & Mentorship Platform

Problem: Slow reviews let bugs through juniors lack guidance seniors waste time on repetitive feedback.

Description: Auto-prioritizes reviews by risk, suggests AI fixes, routes complex changes to experts. Matches juniors with mentors based on feedback patterns.

2. Real-Time Disaster Response System

Problem: Victims can't find aid, volunteers don't know where to go, NGOs duplicate efforts during the critical 72-hour window.

Description: Unified dashboard mapping needs against resources, optimizing supply distribution, preventing gaps through shared situational awareness.

3. Smart Campus Admin Portal

Problem: Students wait blindly for requests; administrators lose track across departments; manual follow-ups waste time.

Description: Centralized portal with real-time status tracking, workflow dashboards, auto-notifications, and escalation rules for delays.

4. Decentralized Credential Verification

Problem: Fake degrees cost millions genuine candidates wait weeks records trapped in silos.

Description: Blockchain-issued tamper-proof credentials with instant employer verification, AI forgery detection, and user-controlled privacy.

BLOCKCHAIN

1. Cross-Chain Asset Bridge

Problem: Create a decentralized bridge to transfer tokens between blockchains without central authority.

Description: Build a trustless system where validators confirm cross-chain transfers with a 7-day fraud challenge period. Uses cryptography and economic incentives to protect funds even if some validators act maliciously.

2. Carbon Credit Trading Platform

Problem: Develop a blockchain marketplace for fraud-free carbon credit trading.

Description: Tokenize carbon credits as unique, traceable digital assets linked to IoT-verified real-world CO₂ reductions. Ensures each physical ton maps to exactly one token—eliminating double-counting and fake credits.

3. Verifiable AI Investment Manager

Problem: Build a transparent AI agent that executes and verifies trades on-chain.

Description: An autonomous system that follows natural language investment rules, manages DeFi portfolios, and records every decision on-chain for full auditability. Includes emergency stop mechanisms for volatile markets.

4. Private Decentralized Exchange (Dark Pool)

Problem: Design a fully private trading platform with encrypted order books.

Description: A decentralized exchange where institutional orders stay encrypted during matching using zero-knowledge proofs. Settlement is public but order details remain hidden—preventing front-running and information leaks.

CYBERSECURITY

1. Fraud Transaction Detection System

Problem: Digital payment growth has spiked unauthorized transactions; weak detection systems fail users.

Description: Real-time analysis of amount, time, location, and frequency patterns to flag suspicious activity, assign risk scores, and alert users while minimizing false positives.

2. Fake News Detection System

Problem: Misinformation spreads rapidly with serious social and political consequences; verification tools are scarce.

Description: AI-powered classification of articles, headlines, and URLs using NLP techniques, delivering authenticity scores and explanations without bias across content types.

3. Secure File Sharing System

Problem: Sensitive file transfers risk unauthorized access, data leaks, and tampering.

Description: Encrypted platform with user authentication, pre-upload encryption, permission-based downloads, and time-limited or one-time access links privacy without performance loss.

4. Data Leak Prevention System (DLP)

Problem: Personal data, passwords, and financial details leak unintentionally or maliciously.

Description: Monitors uploads and inputs for sensitive patterns (emails, Aadhaar, PAN, passwords), blocks or warns before leakage, maintains logs, and preserves normal workflow.

APP DEVELOPMENT

1. Universal Government Form Filler

Problem: Applying for benefits means repeating the same info endlessly one error delays everything 6 months.

Description: Learns from your first form, auto-fills the rest, catches errors pre-submission, keeps answers consistent across agencies.

2. Memory-First Personal Journal

Problem: Social media buries your memories; you scroll instead of remembering.

Description: Uses spaced repetition to resurface your content when you'll remember it, shows life context from that time, slows posting to share what you'll actually want back.

3. Research Fraud Detection System

Problem: Fake research wastes billions and slows real discovery fraud caught too late.

Description: Auto-scans papers for suspicious numbers, maps citation rings, checks author writing styles, rewards scientists who replicate work.

4. Creator-Backer Trust Platform

Problem: Creators take your money and vanish backers have zero recourse.

Description: Escrow holds funds until milestones hit, creators stake reputation, social proof required for higher tiers, abandoned projects become completion bounties.

OPEN INNOVATION

Teams are free to choose any problem statement of their interest. However, the final solution must be built using the prescribed technology stack. This ensures a fair evaluation process while still allowing participants the flexibility to explore innovative and creative ideas in their problem selection.

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